

# For Loops in Python

## Known Repetition, Collections and Nested Loops

Programming Fundamentals Team

SETU

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# Outline

**1. Learning outcomes**

**2. Why another loop?**

**3. For loop basics**

**4. Using range()**

**5. Accumulators with for loops**

**6. Iterating through strings**

**7. Iterating through lists**

**8. Nested loops**

**9. Pattern examples**

**10. Common mistakes**

**11. Bringing it together**

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# 1.1 By the end of this lecture

You should be able to:

- Explain when a for loop is useful
- Use `range()` with start, stop and step values
- Iterate through strings
- Iterate through lists
- Use selection inside a loop
- Use accumulators with for loops
- Write nested loops
- Create simple text patterns

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## 2.1 Review: while loop

```
i = 1

while i <= 10:
    print(i)
    i += 1
```

Use `while` when we may not know how many repetitions are needed.

## 2.2 A simpler way when we know the count

```
for i in range(1, 11):  
    print(i)
```

A for loop is useful when we know what sequence we want to process.

## 2.3 While versus for

### while loop

Use when repetition depends on a condition.

Examples:

- password validation
- menu loops
- sentinel loops

### for loop

Use when processing a sequence.

Examples:

- numbers in a range
- characters in a string
- items in a list



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## 3.1 Syntax

```
for variable in sequence:  
    statements
```

Examples of sequences:

- `range(5)`
- a string such as `"Python"`
- a list such as `[70, 80, 90]`

## 3.2 First for loop

```
for i in range(5):  
    print(i)
```

Output:

```
0  
1  
2  
3  
4
```

## 3.3 Important: range stops before the end

```
range(5)
```

Produces:

0, 1, 2, 3, 4

It does not include 5.

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## 4.1 Start and stop

```
for i in range(1, 11):  
    print(i)
```

This prints 1 to 10.

`range(start, stop)` includes the start but excludes the stop.

## 4.2 Step value

```
for i in range(0, 21, 2):  
    print(i)
```

This prints even numbers from 0 to 20.

## 4.3 Counting backwards

```
for i in range(10, 0, -1):  
    print(i)
```

This prints 10 down to 1.



## 4.4 Activity: predict the output

```
for i in range(2, 12, 3):  
    print(i)
```

Questions:

- What is the first value?
- What is added each time?
- Why is 12 not printed?

## 4.5 Quick challenges

Write for loops to print:

1. 1 to 20
2. 10 to 1
3. Multiples of 5 from 5 to 50
4. Odd numbers from 1 to 19

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## 5.1 Sum numbers 1 to 100

```
total = 0

for i in range(1, 101):
    total += i

print(total)
```

## 5.2 Calculate an average

```
total = 0

for i in range(5):
    mark = int(input("Enter mark: "))
    total += mark

average = total / 5
print("Average:", average)
```

## 5.3 Add validation

```
total = 0

for i in range(5):
    mark = int(input("Enter mark: "))

    while mark < 0 or mark > 100:
        mark = int(input("Invalid. Try again: "))

    total += mark

print("Average:", total / 5)
```

This combines for, while, and or.

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## 6.1 Strings are sequences

```
for letter in "Python":  
    print(letter)
```

Output:

P  
y  
t  
h  
o  
n



## 6.2 Count vowels

```
word = input("Enter word: ")
vowels = 0

for letter in word:
    if letter in "aeiouAEIOU":
        vowels += 1

print("Vowels:", vowels)
```

## 6.3 Activity: count spaces

Write a program that asks for a sentence and counts how many spaces it contains.

Hint:

```
if character == " ":
```

## 6.4 Solution: count spaces

```
sentence = input("Enter a sentence: ")

space_count = 0

for character in sentence:
    if character == " ":
        space_count += 1

print("Number of spaces:", space_count)
```

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## 7.1 What is a list?

A list stores multiple values in one variable.

```
marks = [75, 60, 80, 90]
```

**A for loop is ideal for processing every item in a list.**

## 7.2 Print each mark

```
marks = [75, 60, 80, 90]
```

```
for mark in marks:  
    print(mark)
```

## 7.3 Total and average

```
marks = [75, 60, 80, 90]  
total = 0
```

```
for mark in marks:  
    total += mark
```

```
average = total / len(marks)  
print("Average:", average)
```

## 7.4 Find the highest mark

```
marks = [75, 60, 80, 90]
highest = marks[0]

for mark in marks:
    if mark > highest:
        highest = mark

print("Highest:", highest)
```



## 7.5 Count passes

```
marks = [75, 40, 32, 81, 55]  
passes = 0
```

```
for mark in marks:  
    if mark >= 40:  
        passes += 1
```

```
print("Passes:", passes)
```

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## 8.1 What is a nested loop?

A nested loop is a loop inside another loop.

Outer loop controls the main repetition.

Inner loop completes all of its repetitions for each one repetition of the outer loop.

## 8.2 Rows and Seats Analogy

### Imagine a classroom:

- 3 rows
- 4 seats in each row

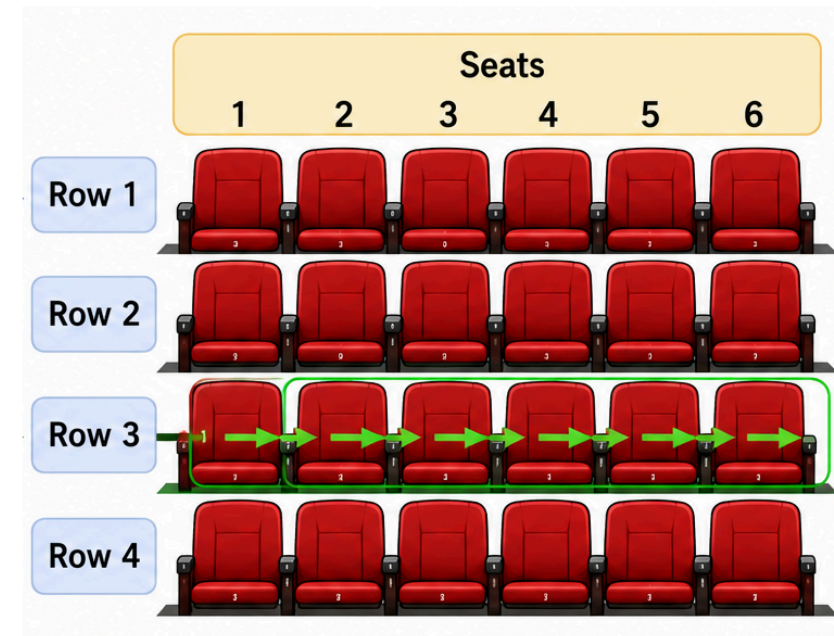
For every row, check every seat.

The **outer loop** selects the row.

The **inner loop** selects each seat in that row.

## Rows and Seats Analogy

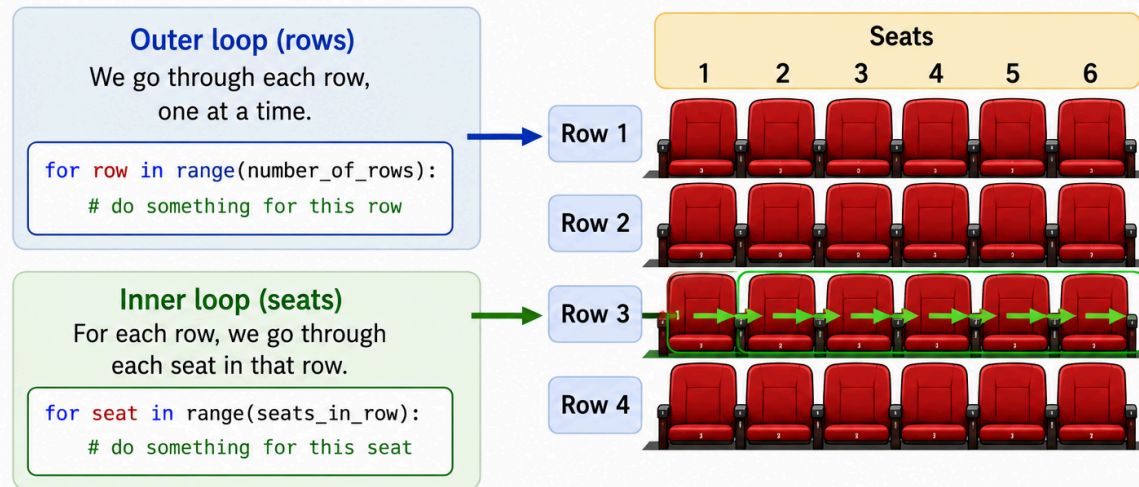
Think of a theatre with rows of seats.



## 8.3 Rows and Seats Analogy Continued

### Rows and Seats Analogy

Think of a theatre with rows of seats.



The **outer loop** chooses the **row**.  
The **inner loop** goes through each **seat** in that row.  
Together, they visit every seat in the theatre.

## 8.4 Rows and Seats Analogy Code

```
for row in range(4):  
    for seat in range(6):  
        print(row, seat)
```

## 8.5 Trace the idea

```
row 0: seat 0, seat 1, seat 2, seat 3, seat 4, seat 5
row 1: seat 0, seat 1, seat 2, seat 3, seat 4, seat 5
row 2: seat 0, seat 1, seat 2, seat 3, seat 4, seat 5
row 3: seat 0, seat 1, seat 2, seat 3, seat 4, seat 6
```

The inner loop runs completely for each value of row.

## 8.6 Is this code better?

```
for row in range(1,5):  
    print("\nRow ", row, end=' ')  
    for seat in range(1,7):  
        print(" Seat ", seat, end=' ')
```



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## 9.1 Print a square

```
for row in range(5):  
    print("*" * 5)
```

Output:

```
*****  
*****  
*****  
*****  
*****
```

## 9.2 Print a triangle

```
for row in range(1, 6):  
    print("*" * row)
```

Output:

```
*  
**  
***  
****  
*****
```

## 9.3 Multiplication table

```
for i in range(1, 11):  
    print(i, "x 7 =", i * 7)
```

## 9.4 1 to 10 tables

```
for table in range(1, 11):  
    print("Table", table)  
  
for i in range(1, 11):  
    print(i, "x", table, "=", i * table)  
  
print()
```

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## 10.1 Off-by-one errors

```
for i in range(1, 10):  
    print(i)
```

This prints 1 to 9, not 1 to 10.

**The stop value is excluded.**

## 10.2 Indentation errors

```
for i in range(5):  
    print(i)
```

Python needs indentation to show what belongs inside the loop.



## 10.3 Changing the wrong variable

```
total = 0
```

```
for mark in [50, 60, 70]:  
    total = mark
```

This replaces the total each time.

Better:

```
total += mark
```

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## 11.1 Mini-program: mark report

```
marks = [72, 38, 55, 90, 41]

total = 0
passes = 0
highest = marks[0]

for mark in marks:
    total += mark

    if mark >= 40:
        passes += 1

    if mark > highest:
        highest = mark

print("Average:", total / len(marks))
print("Passes:", passes)
print("Highest:", highest)
```

## 11.2 What does this combine?

- Lists
- for loops
- Selection
- Counters
- Accumulators
- Highest value pattern
- len() function

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## 12.1 Today we learned

- Use for loops to process a sequence.
- `range()` generates numbers.
- The stop value in `range()` is excluded.
- Strings and lists can be looped through directly.
- Selection can be placed inside loops.
- Nested loops are useful for grids, tables and patterns.

## 12.2 Practice exercises

1. Print numbers from 1 to 50.
2. Print multiples of 3 from 3 to 60.
3. Print each character in your name.
4. Count vowels in a sentence.
5. Calculate the total of a list of numbers.
6. Count how many marks are passes.
7. Find the highest number in a list.
8. Print a 5 by 5 square of stars.
9. Print a right-angled triangle.
10. Challenge: print multiplication tables from 1 to 10.

# Thanks for Watching - Any questions?

